

Robotics and Artificial Intelligence Nigeria (RAIN)

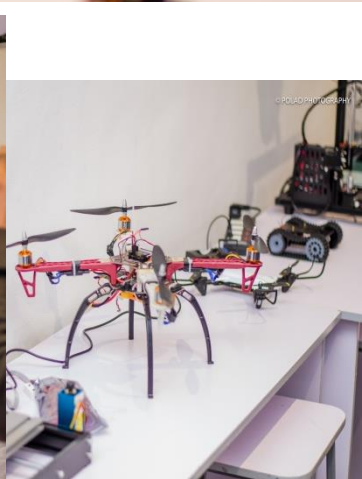
By Dr Olusola Ayoola

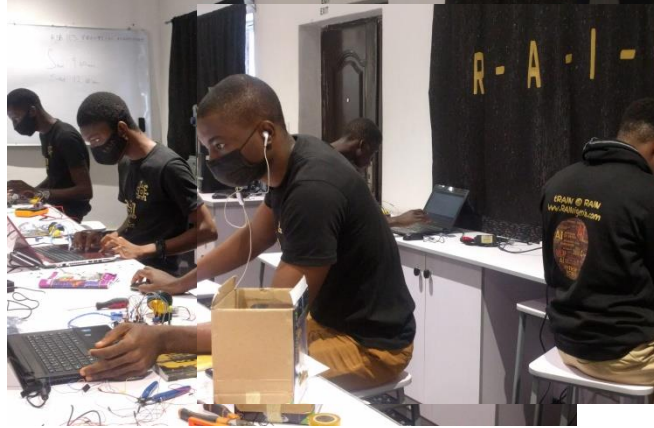
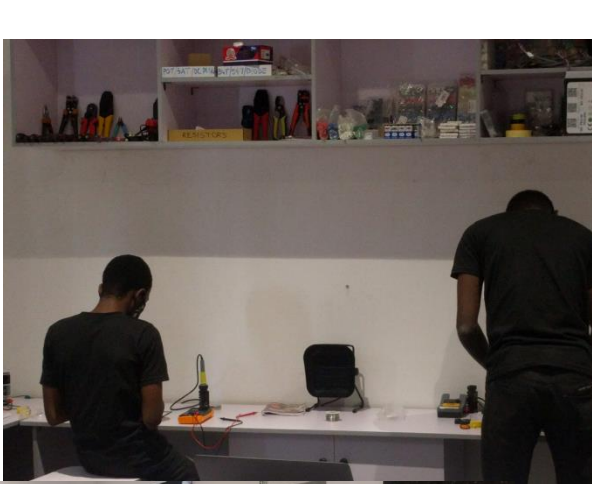
Partnership for Technological Advancement

Dr. Olusola Sayeed Ayoola, B.Sc (Ib.) M.Sc (Manc.), Ph.D (Manc.), MIET ,MNSE,MNCS
CEO | Robotics and Artificial Intelligence Nigeria (RAIN)

RAIN

- Founded in 2019
- Robotics and Artificial Intelligence Nigeria (RAIN) has grown to be Nigeria's leading facility for research, training and certification in Robotics and Artificial Intelligence.
- Located in a serene environment in Iyaganku G.R.A, Ibadan, Oyo State.
- RAIN designs a structured training curriculum, delivers training, conducts research and engages in product designs and development for organizations.





RAIN: The Facility is

- Home to a fully equipped robotics laboratory, sound academics and researchers.
- Nigeria's concentration hub of highly skilled software and hardware developers that see no impossibilities.
- Leading the discussion with stakeholders across sectors on Artificial Intelligence (AI) based interventions.





RAIN: The Objective

- Develop a robust Curriculum in Robotics and AI.
- Develop a large number of **skilled** workforce through practical trainings with live projects.
- Enhance industry growth through 4IR-based interventions.
- Promote technological advancement through ground-breaking research in AI and Robotics.
- Encourage wide-spread adoption of AI and R.

RAIN: The Strength

- Proven competitive advantage
- Product Design (3D) & Parts Fabrication
- Internet of Things (IOT)
- Advanced Machine Learning
- Deep Neural Networks
- Computer Vision and Natural Language Processing NLP
- Simultaneous Localization and Mapping (SLAM)
- Robot Operated Systems (ROS)
- Robot Process Automation (RPA)
- Unmanned Aerial, Ground and Underwater Systems
- Real-World Model Deployment
- App Development (Web, Desktop, Android & iOS)

RAIN: The Milestones

- Robotics and Artificial Intelligence Nigeria (RAIN) was founded in 2019 and has since grown to be Nigeria's leading facility for research, training and certification in Robotics and Artificial Intelligence.
- First International Student Arrival – October 2022
- Visit of the Deputy Secretary General of the United Nations – December 2021
- Resolution by the House of Representatives tasking the Federal Government of Nigeria to liaise with RAIN – December 2021



At Robotics and Artificial Intelligence Nigeria (R-A-I-N)

RAIN: The Milestones

- The Federal Government has collaborated with RAIN severally through the following organs:
 - National Information Technology Development Agency (NITDA)
 - Security Departments
 - Federal Ministry of Youths and Sport Development
- Other Collaborators include:
 - MIT LaunchX 2022
 - Nile University, Egypt AI4Dev 2022
 - The Government of Kogi State
 - The United Nations Development Programme (UNDP)
 - Adeus Ltd, United Kingdom
 - One Watt Solar (OWS) Ltd, UK
 - Trade & Investment Consultants Ltd, UK



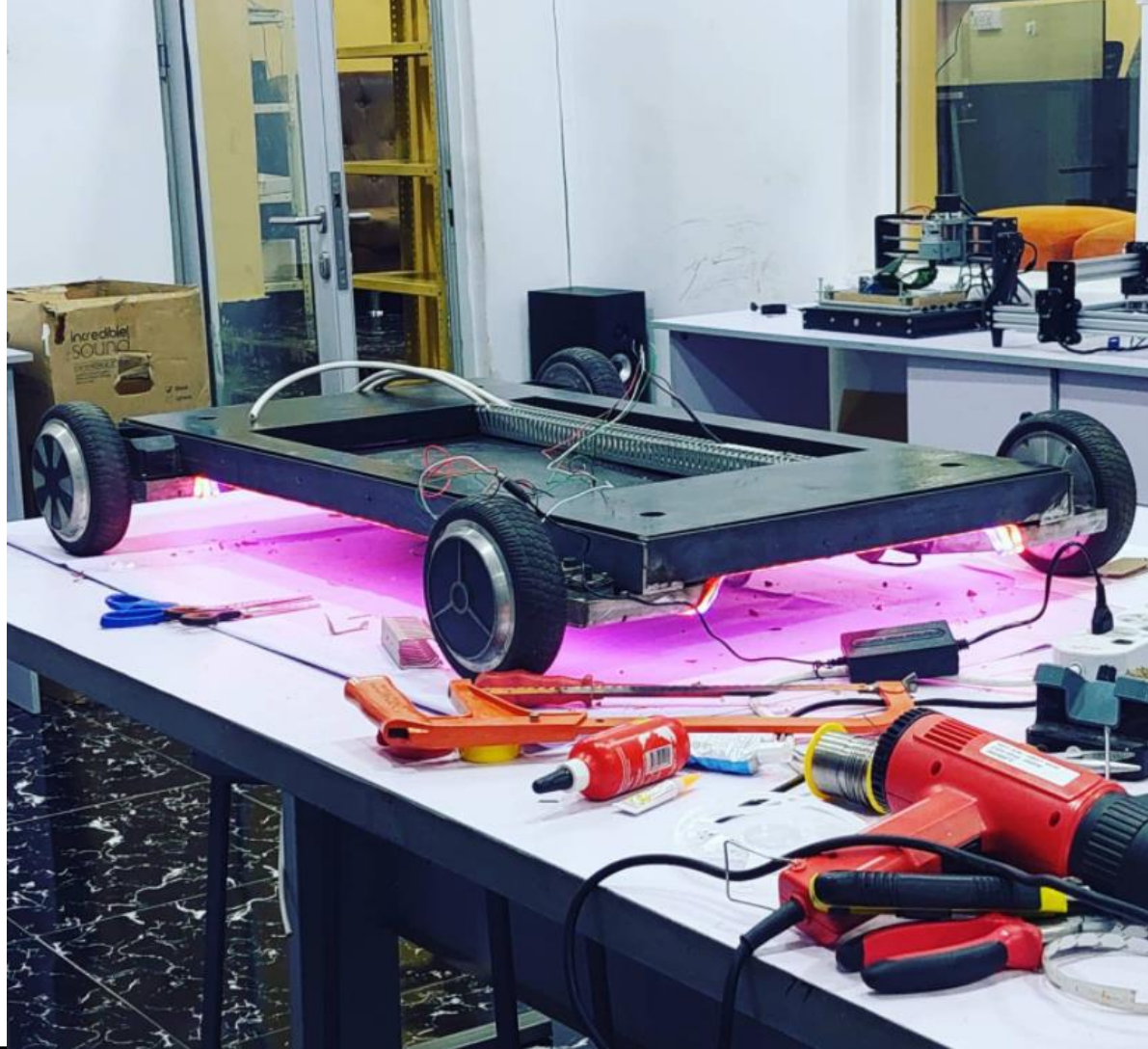
Collaboration with Nigeria Police (Oyo State)



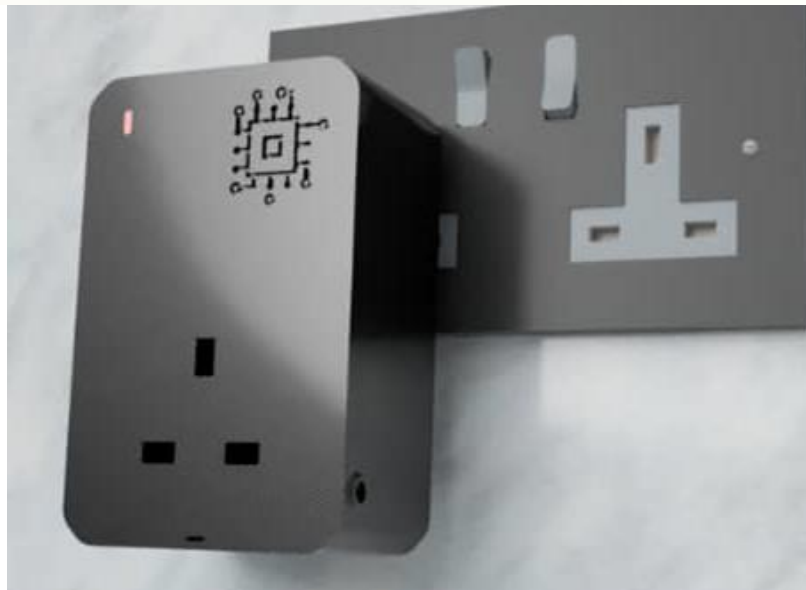
Collaboration with Raspberry Pi Africa Region

RAIN: Notable Projects

1. RAIN CEDBAS - CARBON EMISSION BASED AERIAL SURVEILLANCE SYSTEM
2. RAIN DBCV – AI-CV Based Drone Footage Analysis for Human Detection
3. RAIN IMoRP - Industrial Mobile Robotic Platform
4. RAIN SOS -90 - SMS and IOT based emergency GPS tracking and alarm system for personal protection
5. RAINY: IOT and speech controlled smart socket and power metering system for apartments
6. RAIN-PAS: IOT based public address system for information dissemination
7. RAIN Physio-01: IOT based robotic sleeve for physiotherapy exercises and wearable sensors for analytics
8. RAIN SC-75: AI-IOT based crop farm irrigation control system
9. RAIN SC-76: AI-IOT based poultry farm brooding temperature control system
10. Mobile traffic light solution
11. Invisible perimeter fencing solution



RAIN IMoRP at the early stage



The RAIN SC-75/76 Device



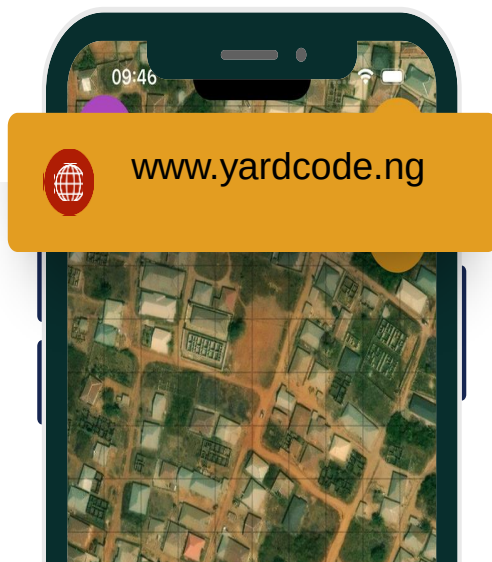
RAIN: Other Potentials

- City Wide Central Surveillance System
- Robust Unmanned Ground Vehicle for Extreme Environment Exploration
- Training to Solve the Existing Skill Gap



YARDCODE

DIGITAL ADDRESSING FOR
NIGERIA



INTRODUCING YARDCODE



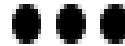
FREE TO USE

No hidden charges. Every Nigerian gets their YardCode at zero cost - forever



WORKS ANYWHERE

Works 24/7, rain or shine. When other systems fail, YardCode delivers



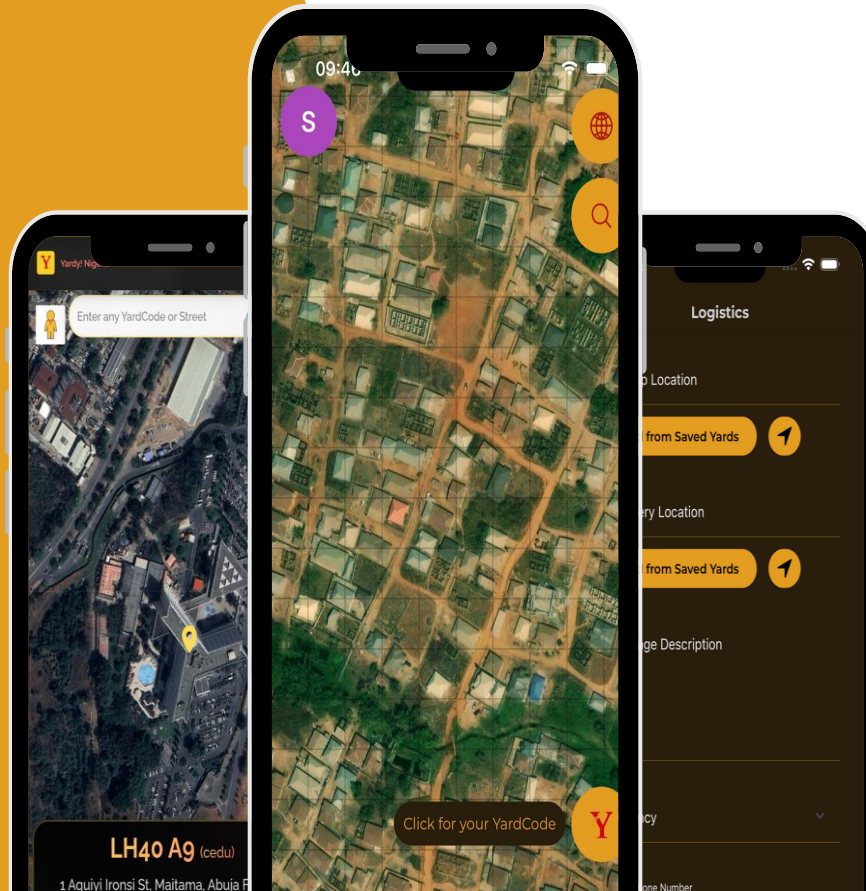
RELIABLE

From Lagos Island to remote villages - if you're in Nigeria, you can get your YardCode



PERFECT FOR YOU

Simple enough for grandma, smart enough for tech experts. Built for every Nigerian



THE PROBLEM WE'RE SOLVING

CURRENT CHALLENGES:



Current postcode system is obsolete
and fails to meet digital needs



Inconsistent Address Standards
No unified system for house numbering and street naming



Infrastructure Gaps
Remote areas lack proper addressing, hindering service delivery



Economic Impact
Poor addressing limits financial inclusion, logistics, and emergency services



GPS Complexity
Traditional GPS coordinates are too long and difficult to remember

COST:



Delayed emergency responses



Failed package deliveries



Poor census and demographic data



Lack of Brand Differentiation



Reduced investment in remote areas

What Sets YardCode Apart

UNIQUE IDENTIFIER

Every 1m x 1m grid gets a unique, memorable code (e.g., GQ9 U88)

MATHEMATICAL FOUNDATION

No database dependency - purely mathematical relationships

RELATED POSITIONING

Adjacent locations have related codes (unlike What3Words)

PERMANENT

SYSTEM Sustainably energy-efficient, requiring minimal computational power

SMALLER GRID

1m x 1m precision vs. competitors' 3m x 3m

NO DATABASE DEPENDENCY

Mathematical algorithm reduces infrastructure needs

LOGICAL SEQUENCE

Related codes for adjacent areas

NIGERIAN-BUILT

Designed specifically for Nigerian addressing standards

Key Differentiators

COMPREHENSIVE PLATFORM SERVICES

Beyond Just Codes - A Complete Ecosystem*

WEB SERVICE



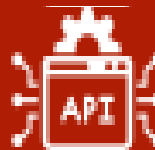
- Free YardCode lookup for all Nigerians
 - Street address to YardCode conversion
 - Google Street View integration for accuracy
 - View and share local yard gossip

MOBILE APP

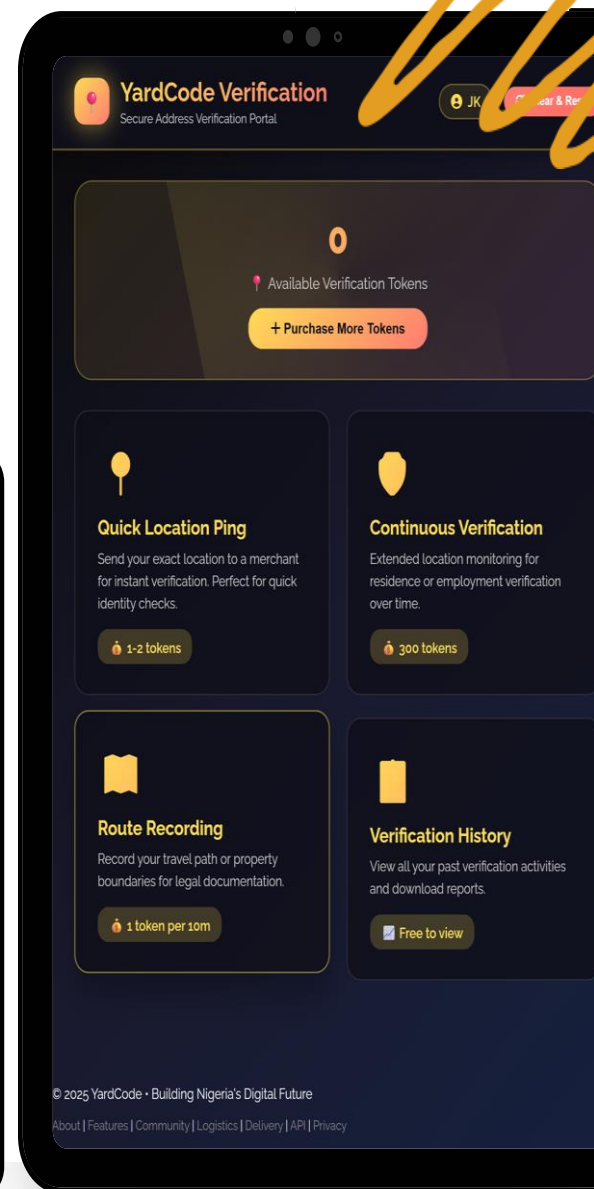


- Offline functionality-works without internet
- Available on Google Play Store and Apple App Store
- Real-time location services

DEVELOPER API



- Convert YardCodes to GPS coordinates
 - Integration with existing platforms
- Scalable for enterprise solutions



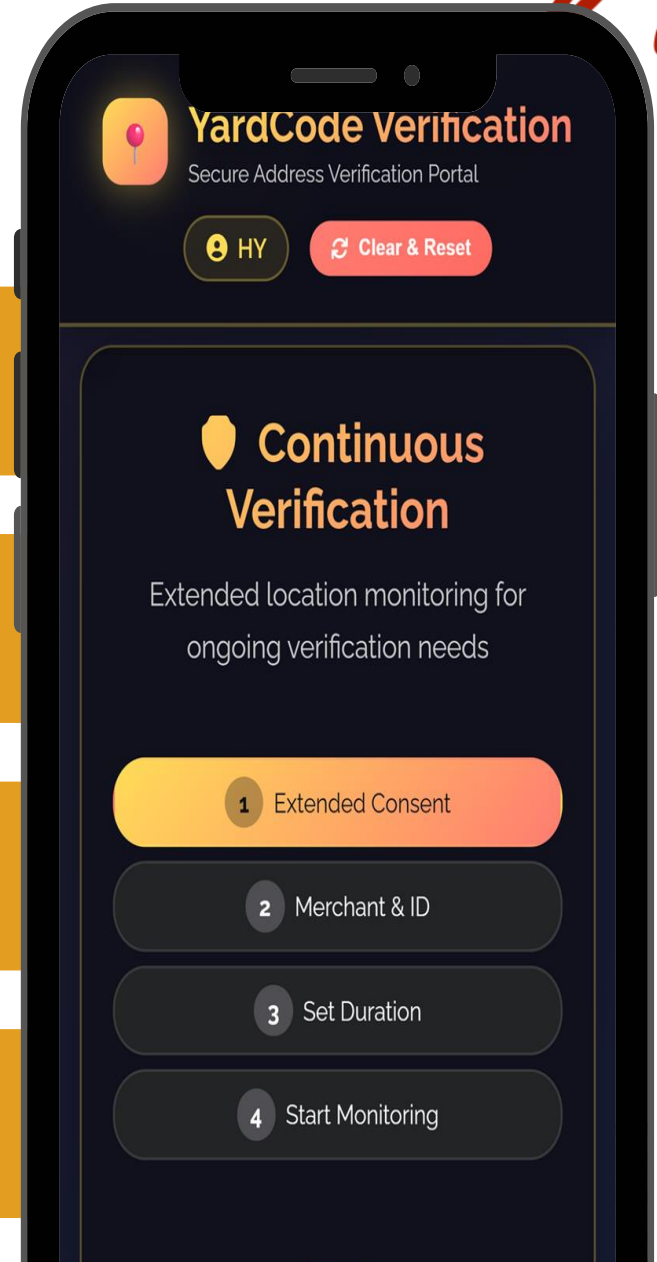
ADDRESS VERIFICATION PORTAL

01 AI-powered residency verification for over 7-30 days

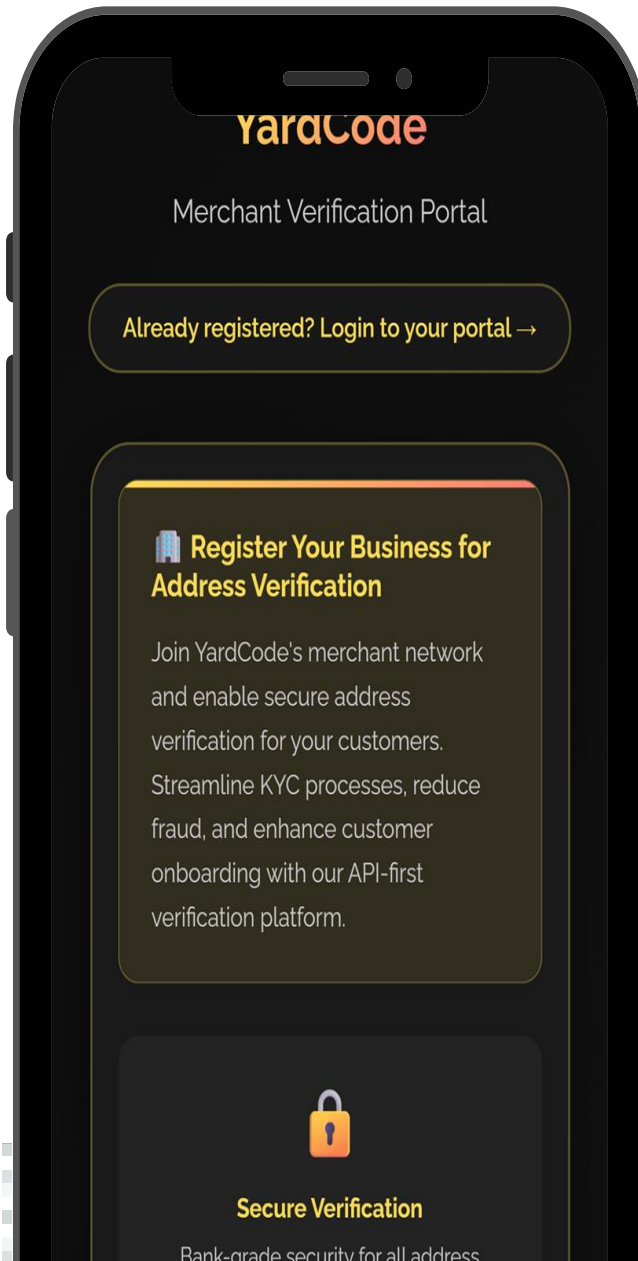
02 28 daily location pings for accuracy

03 20-meter tolerance for same premise verification

04 Digital certificates for confirmed addresses & Annual re-verification reminders



MERCHANT PORTAL



01

Dashboard for managing location requests

02

User permission management

03

Billing/subscription management

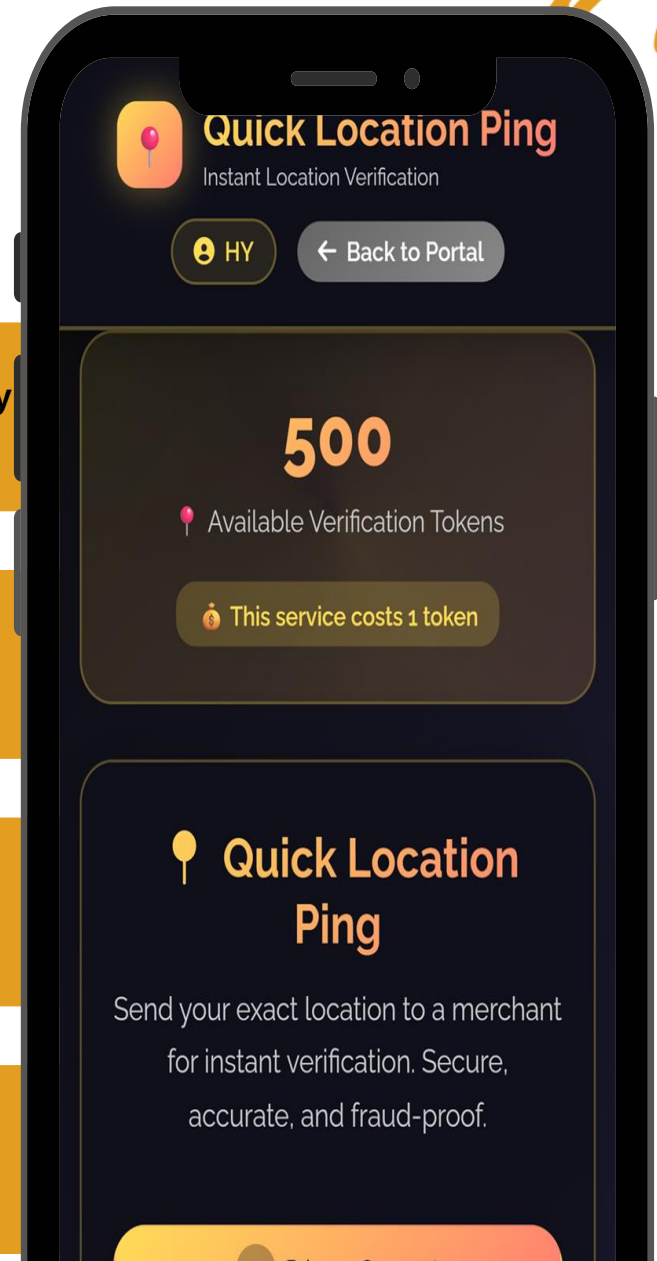
PING ADDRESS FEATURE

01 Uneditable location sharing for security and trust

02 Instant YardCode + photo verification

03 Perfect for field workers, students, security personnel

04 Direct integration via email or websocket



ROUTE RECORDING



01 **Start-to-end route tracking**

02 **Automatic RouteCode generation**

03 **Organizational integration for logistics**

THE FUTURE IS HERE

YARDCODE ISN'T JUST
SOLVING A PROBLEM – WE'RE
BUILDING THE FOUNDATION
FOR NIGERIA'S DIGITAL
FUTURE

EVERY ADDRESS MATTERS.
EVERY LOCATION COUNTS.
*WELCOME TO THE YARDCODE
ERA*



NRS is actually the Engine Room for National Prosperity

Reimagining the Nigeria Revenue Service as Nigeria's growth partner

THE IMPORTANCE OF AI

Copyright Violation Detection: Use AI tools to monitor digital platforms for pirated content automatically.

Document Review and Automation: NLP tools to quickly analyse contracts, case laws, and submissions.

Chatbots for Public Inquiries: Provide 24/7 assistance to stakeholders about copyright laws and processes.

Automatic copyright registration: AI checks for existing duplicates in the NCC database.

Administrative Efficiency: Data analysis for strategic decision-making

WHAT MORE?

Three Platforms Where Growth Meets Government

The Nigerian Trade Exchange

An Alibaba-style marketplace for Nigerian goods and services to reach Africa and the world.

- Nigerian manufacturers, artisans, farmers, and service providers sell across Africa and globally
- Integrated payment rails and settlement, in partnership with the Central Bank of Nigeria
- Tax calculated and collected automatically at point of sale — compliance without confrontation
- Logistics and delivery coordinated through the platform, order to doorstep
- Every transaction feeds real trade and revenue data back to the Service

PLATFORM TWO

The National Employment & Services Registry

A portal where Nigeria's workers — formal and informal — are seen, connected, and folded into the tax net.

- Freelancers, consultants, and contractors register and are discoverable by clients nationwide
- Employers and job-seekers meet on one portal — tax registration becomes a byproduct of finding work
- PAYE and related remittances automated at the point of payment
- A living record of who is working, where, and how — the raw material for real employment statistics

PLATFORM THREE

The Business Growth Desk

A shift in posture — from an office businesses fear, to a desk businesses visit to grow.

- Tax advisory tailored to a business's actual stage: startup, SME, or scale-up
- A real-time compliance dashboard — business owners always know exactly what they owe
- Pathways connecting compliant businesses to credit, grants, and partnership opportunities
- Tax education delivered in the language and channel Nigerians already use

THE INTELLIGENCE LAYER

What AI Adds to the Picture

Six Ways AI Strengthens the Service

Compliance & Risk Profiling

Flag unusual filing patterns and prioritise audit attention where it's genuinely needed.

Tax Education at Scale

Voice notes, chatbots, and local-language content that meet Nigerians where they are.

Automated Auditing

Pre-audit analysis narrows what a human auditor reviews — humans stay in the loop.

Fraud & Impersonation Detection

Continuous monitoring for cloned payment portals impersonating the Service.

Field Officer Monitoring

YardCode SDK integration tracks officer location, duration, and outcomes in the field.

Public-Private Delivery

RAIN and partners build and operate; government sets policy and shares the benefit.

What RAIN Already Brings to the Table

- 01 Building the People**

AIML and RDA programmes training the talent capable of building and operating platforms like these, at home.
- 02 Building the Tools**

RIDE, a browser-native robotics simulation platform, and YardCode, Nigeria's own geo-addressing infrastructure — real, deployed systems.
- 03 Shaping the Policy**

Co-authorship of Nigeria's National AI Strategy and a seat on the UNDP AI4Dev Reference Group.
- 04 Building Across Sectors**

Work spanning education, security, and public administration — including AI curricula for Meta's AI Academy Nigeria.

SUMMARY

The Proposal in One Table

Dimension	Today	The RAIN Vision
Primary posture	Collector of what is owed	Partner in what can be earned
Relationship with business	Compliance and enforcement	Growth, trade, and market access
Data on citizens	Filed returns	Live employment, trade, and income data
Technology role	Digitised forms and portals	AI-driven compliance, education, fraud prevention, field intelligence
Where revenue comes from	Enforcement pressure	A growing economy, willingly compliant